

```

29.     except NameError:
30.         pass # Do nothing if pyperclip wasn't installed.
31.
32.
33. def englishToLeetspeak(message):
34.     """Convert the English string in message and return leetspeak."""
35.     # Make sure all the keys in `charMapping` are lowercase.
36.     charMapping = {
37.         'a': ['4', '@', '/-\\'], 'c': ['('], 'd': ['|']}, 'e': ['3'],
38.         'f': ['ph'], 'h': ['']-['', '|-|'], 'i': ['1', '!', '|'], 'k': ['<'],
39.         'o': ['0'], 's': ['$'], '5'], 't': ['7', '+'], 'u': ['|_|'],
40.         'v': ['\\|/']}
41.     leetspeak = ''
42.     for char in message: # Check each character:
43.         # There is a 70% chance we change the character to leetspeak.
44.         if char.lower() in charMapping and random.random() <= 0.70:
45.             possibleLeetReplacements = charMapping[char.lower()]
46.             leetReplacement = random.choice(possibleLeetReplacements)
47.             leetspeak = leetspeak + leetReplacement
48.         else:
49.             # Don't translate this character:
50.             leetspeak = leetspeak + char
51.     return leetspeak
52.
53.
54. # If this program was run (instead of imported), run the game:
55. if __name__ == '__main__':
56.     main()

```

After entering the source code and running it a few times, try making experimental changes to it. On your own, you can also try to figure out how to do the following:

- Modify the `charMapping` dictionary so that it supports new leetspeak characters.
- Add a feature that can convert leetspeak back into plain English.

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you change `return leetspeak` on line 51 to `return message`?
2. What happens if you change `char.lower()` on line 44 to `char`?
3. What happens if you change `char.lower()` on line 44 to `char.upper()`?
4. What happens if you change `leetspeak = leetspeak + leetReplacement` on line 47 to `leetspeak = leetReplacement`?